

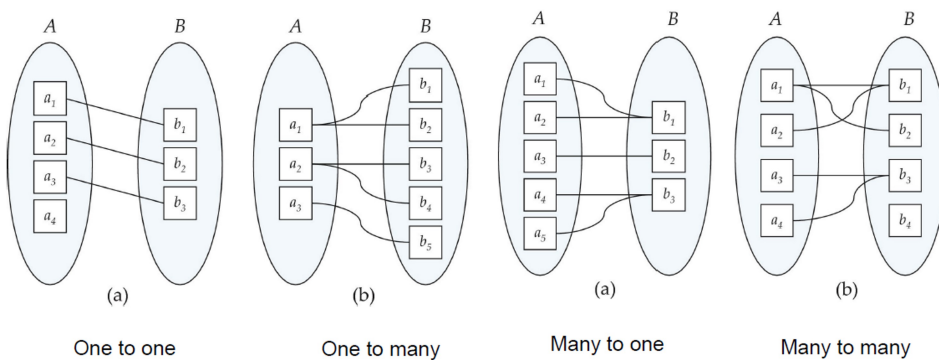
Mapping Cardinalities

Mapping cardinalities, or cardinality ratios, express the number of entities to which another entity can be associated via a relationship set.

Most useful in describing binary relationship sets.

For a binary relationship set the mapping cardinality must be one of the following types:

One to one, One to many, Many to one, Many to many

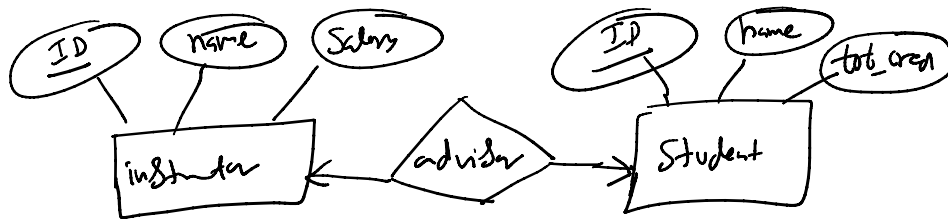
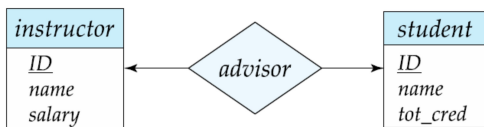


Note: Some elements in A and B may not be mapped to any elements in the other set

We express cardinality constraints by drawing either a directed line ($\rightarrow$), signifying "one," or an undirected line (—), signifying "many," between the relationship set and the entity set.

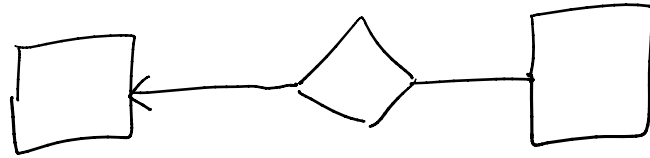
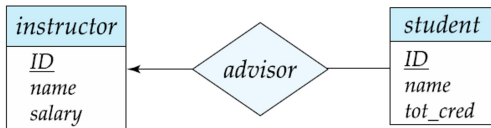
One to one relationship between an instructor and a student:

- An instructor is associated with at most one student via the relationship advisor
- A student is associated with at most one instructor via the relationship advisor



One to many relationship between an instructor and a student:

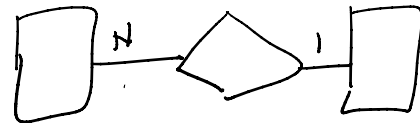
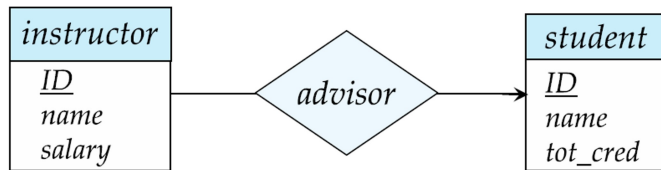
- an instructor is associated with several (including 0) students via advisor
- a student is associated with at most one instructor via advisor



Many-to-One Relationships

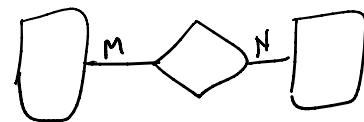
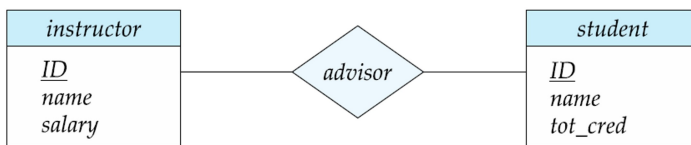
In a many-to-one relationship between an *instructor* and a *student*,

- an *instructor* is associated with at most one *student* via *advisor*,
- and a *student* is associated with several (including 0) *instructors* via *advisor*



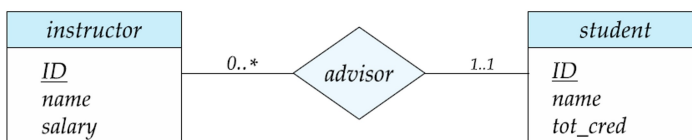
Many-to-Many Relationship

- An *instructor* is associated with several (possibly 0) *students* via *advisor*
- A *student* is associated with several (possibly 0) *instructors* via *advisor*

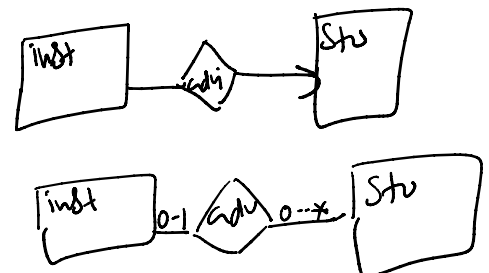


Notation for Expressing More Complex Constraints

- A line may have an associated minimum and maximum cardinality, shown in the form *l..h*, where *l* is the minimum and *h* the maximum cardinality
 - A minimum value of 1 indicates total participation.
 - A maximum value of 1 indicates that the entity participates in at most one relationship
 - A maximum value of * indicates no limit.
- Example



$l \dots h$ $0 \dots 1$
 $2 \dots *$



- Instructor can advise 0 or more students. A student must have 1 advisor; cannot have multiple advisors

The line between advisor and student has a cardinality constraint of 1..1, meaning the minimum and the maximum cardinality are both 1. That is, each student must have exactly one advisor. The limit 0..* on the line between advisor and instructor indicates that an instructor can have zero or more students. Thus, the relationship advisor is one-to-many from instructor to student, and further the participation of student in advisor is total, implying that a student must have an advisor.

It is easy to misinterpret the 0..* on the left edge and think that the relationship advisor is many-to-one from instructor to student—this is exactly the reverse of the correct interpretation.

If both edges have a maximum value of 1, the relationship is one-to-one. If we had specified a cardinality limit of 1..* on the left edge, we would be saying that each instructor must advise at least one student.

Choice of Primary key for Binary Relationship

- Many-to-Many relationships. The preceding union of the primary keys is a minimal superkey and is chosen as the primary key.
- One-to-Many relationships. The primary key of the "Many" side is a minimal superkey and is used as the primary key.
- Many-to-one relationships. The primary key of the "Many" side is a minimal superkey and is used as the primary key.
- One-to-one relationships. The primary key of either one of the participating entity sets forms a minimal superkey, and either one can be chosen as the primary key.

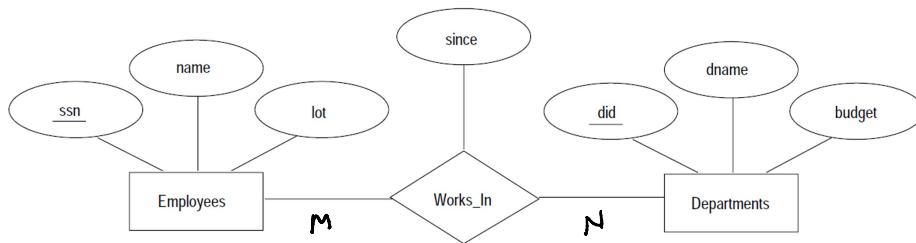


Figure The Works_In Relationship Set

work in

ssn	did	since
e ₁	d ₁	
e ₁	d ₂	
e ₃	d ₁	

PK:

ssn did

many to many

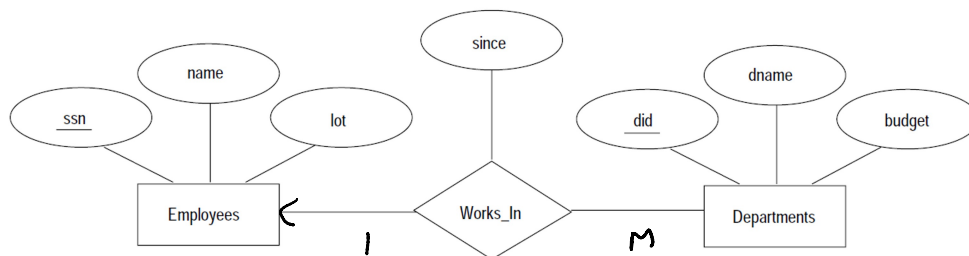
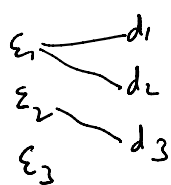


Figure The Works_In Relationship Set



work in

ssn	did	since
e ₁	d ₁	
e ₁	d ₂	
e ₂	d ₃	

PK: did

one to many

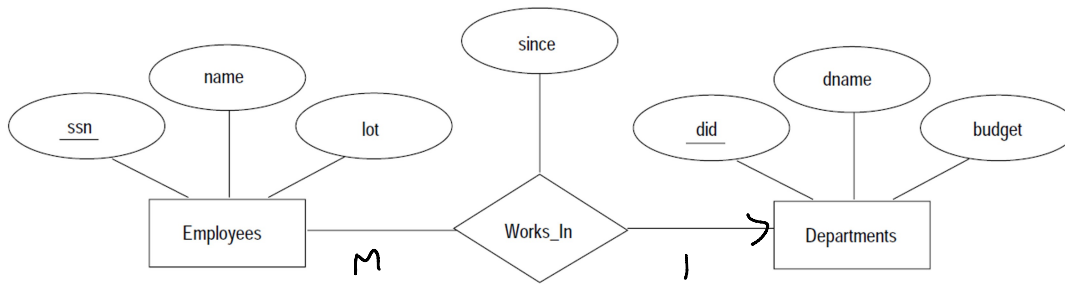


Figure The Works_In Relationship Set

ϵ_1 d_1
 ϵ_2 d_1
 ϵ_3 d_2
 many to one
 work_In ; ssn
 P.K.

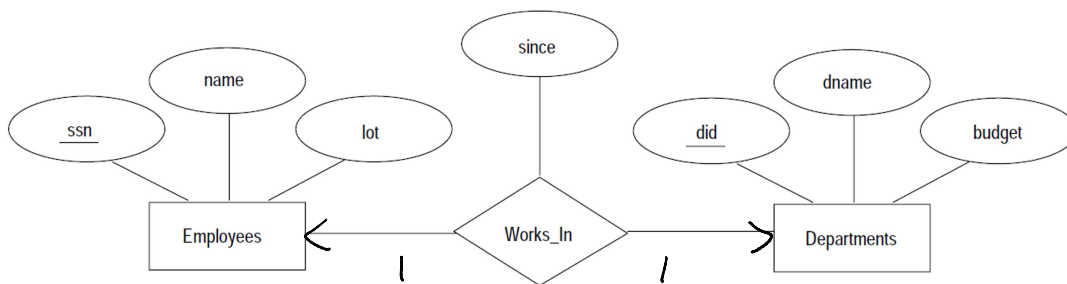


Figure The Works_In Relationship Set

work_In

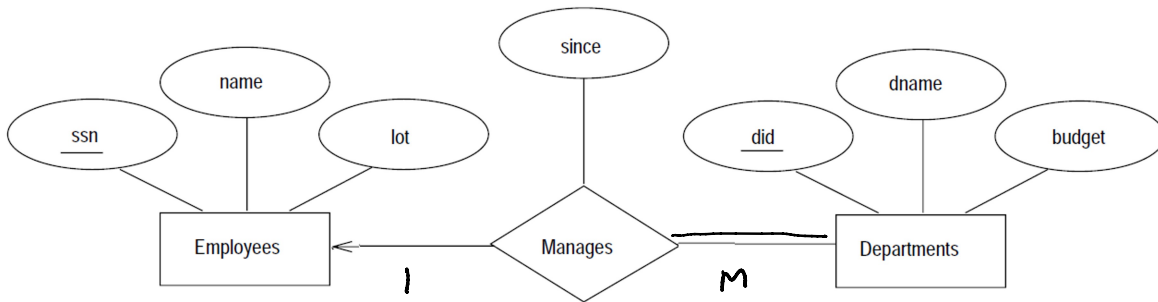
ssn	did	since
ϵ_1	d_1	
ϵ_2	d_3	
ϵ_3	d_2	

one to one

PK: ssn or did

Reduction to Relation Schemas

Consider relationship set called *Manages* between the *Employees* and *Departments* entity sets such that each department has at most one manager, although a single employee is allowed to manage more than one department.



One employee can be associated with many departments (in the capacity of a manager), whereas each department can be associated with at most one employee as its manager.

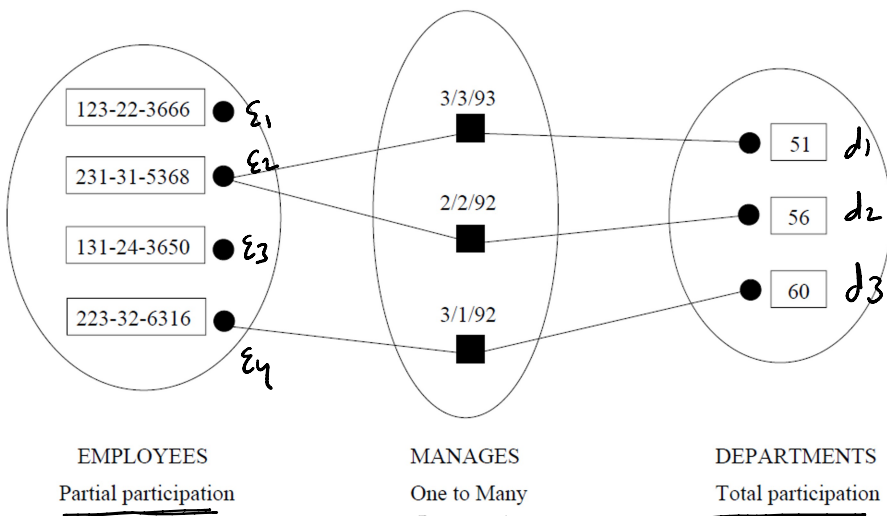
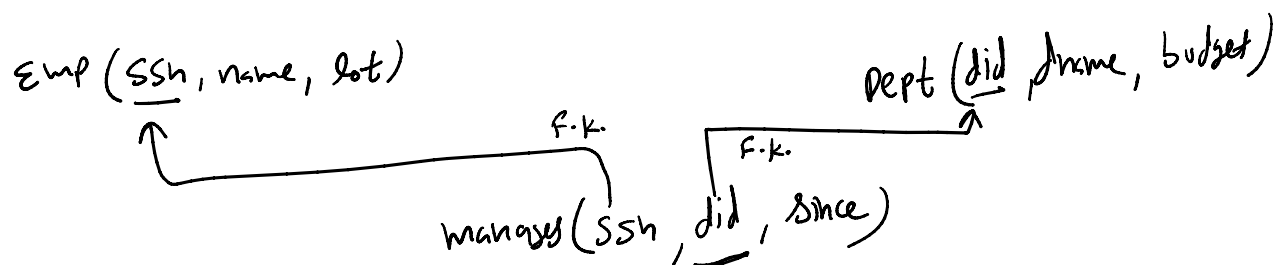
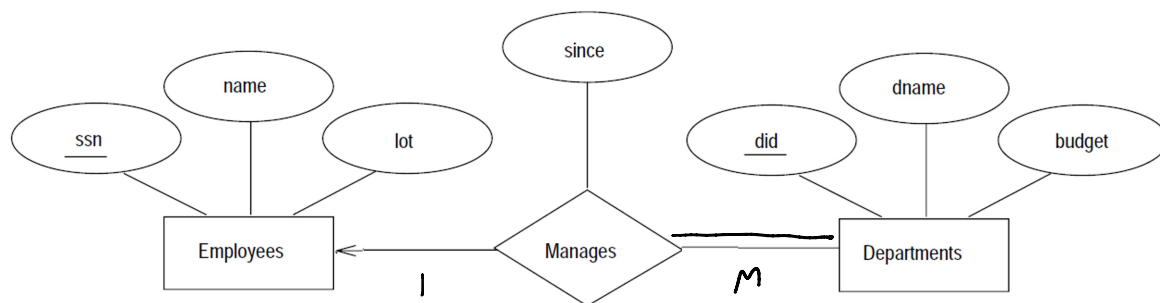


Figure An Instance of the *Manages* Relationship Set

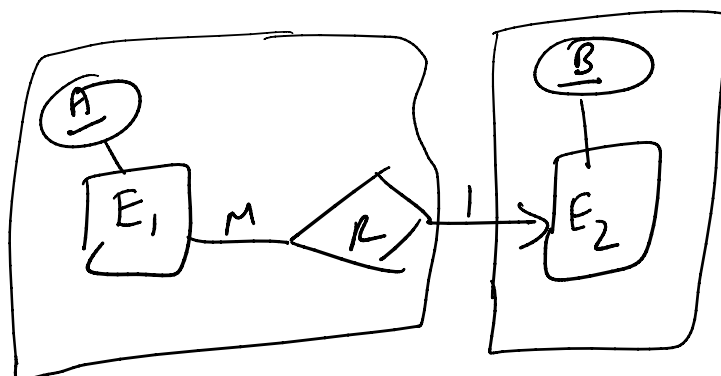
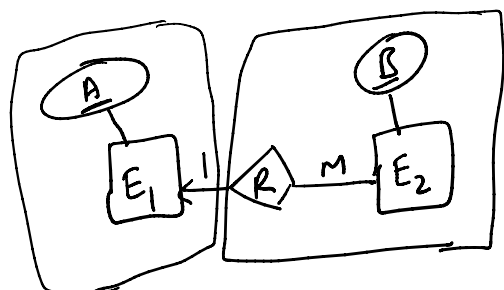
Redundancy of Schemas

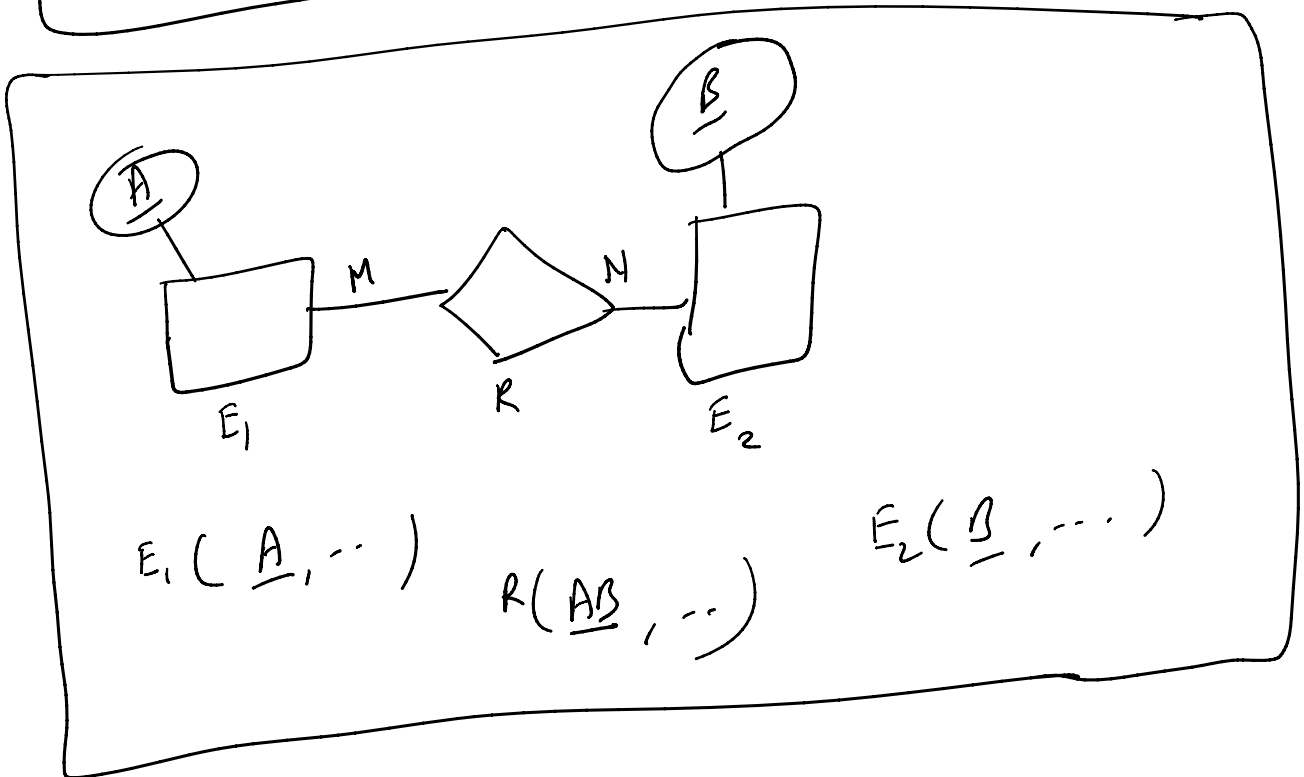
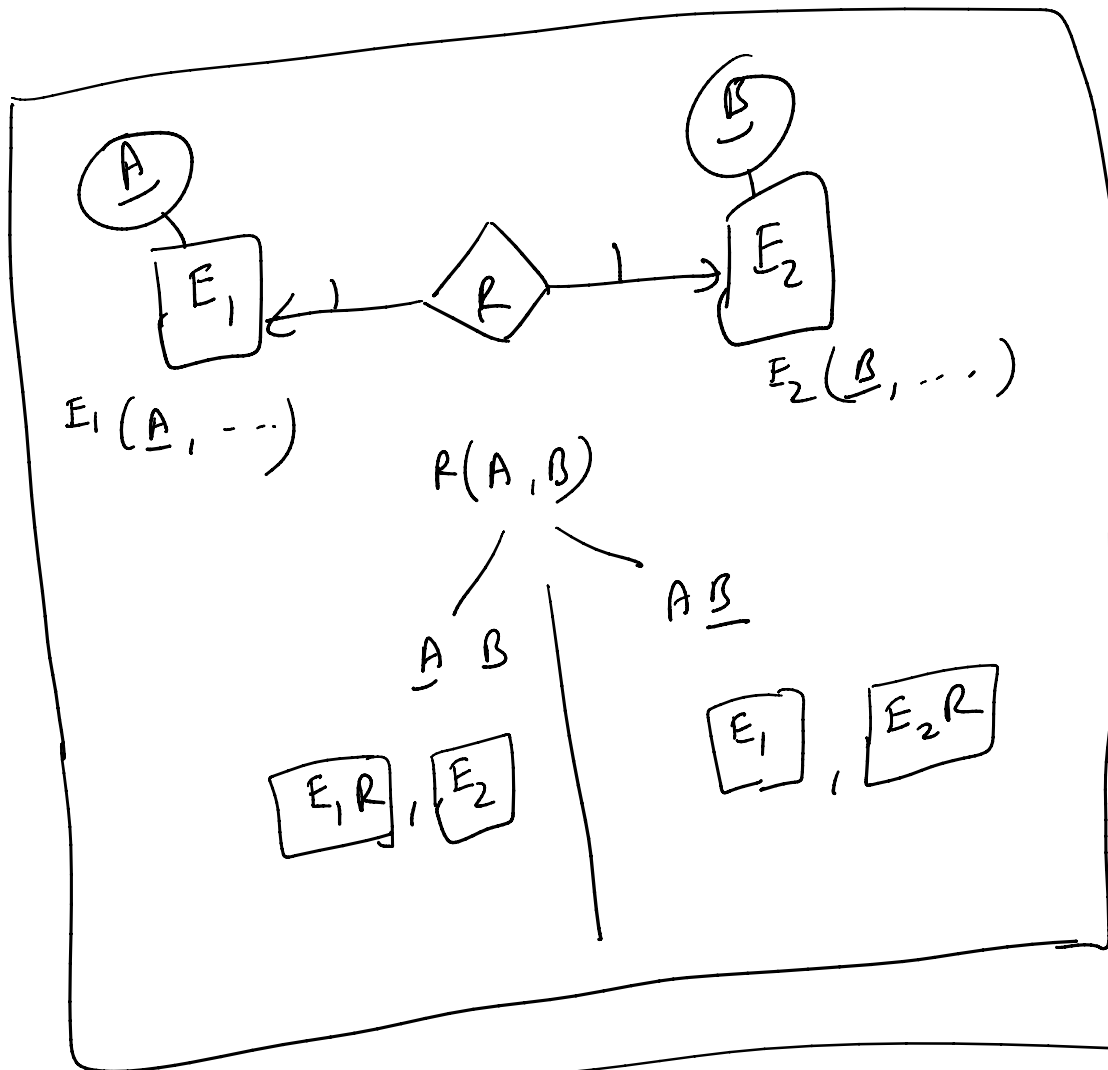


Emp (ssn, name, lot)

Dept-manages (did, dname, budget, ssn, since)

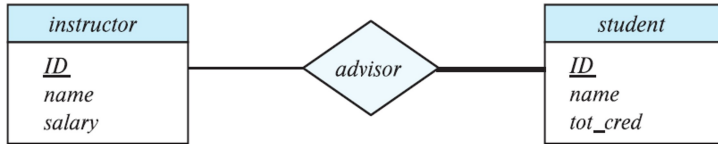
d1	-	-	-	-
d2	-	-	-	-
d3	-	-	NULL	NULL





Total and Partial Participation

- **Total participation** (indicated by double line): every entity in the entity set participates in at least one relationship in the relationship set



participation of *student* in *advisor* relation is total

- every *student* must have an associated instructor
- **Partial participation**: some entities may not participate in any relationship in the relationship set
 - Example: participation of *instructor* in *advisor* is partial

Role indicators:

The entity sets that participate in a relationship set need not be distinct; sometimes a relationship might involve two entities in the same entity set. For example, consider the Reports_To relationship set that is shown in Figure below. Since employees report to other employees, every relationship in Reports_To is of the form (emp1; emp2), where both emp1 and emp2 are entities in Employees.

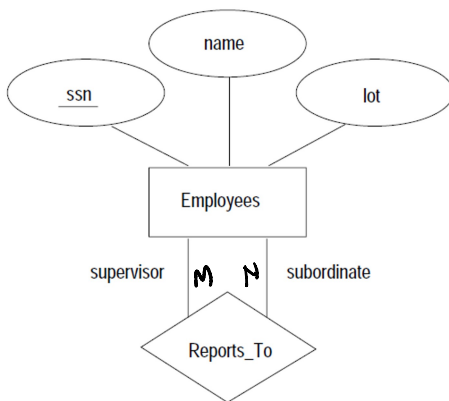
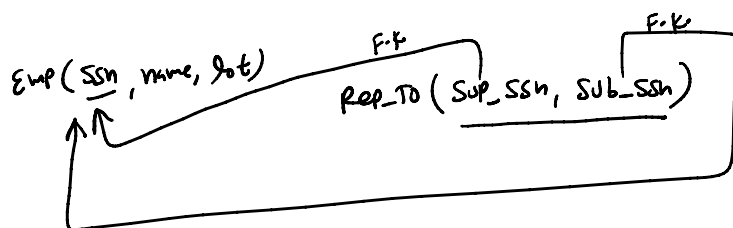


Figure The Reports_To Relationship Set

However, they play different roles: emp1 reports to the managing employee emp2, which is reflected in the role indicators supervisor and subordinate in Figure. If an entity set plays more than one role, the role indicator concatenated with an attribute name from the entity set gives us a unique name for each attribute in the relationship set. For example, the Reports_To relationship set has attributes corresponding to the ssn of the supervisor and the ssn of the subordinate, and the names of these attributes are supervisor_ssn and subordinate_ssn.



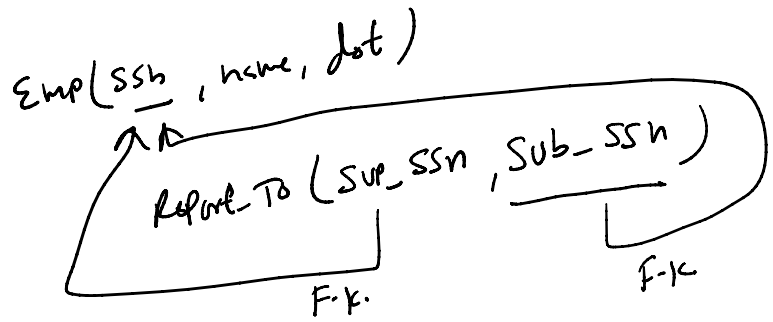
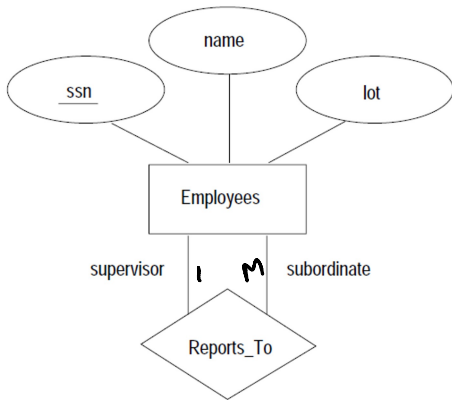


Figure The Reports_To Relationship Set

$Emp_Report_to(name, lot, sup_ssn, \underline{ssn})$